

Answer **all** questions.

1. Cystic fibrosis is a genetic condition that affects the lungs and other organs. It is caused by a mutation.

(a) State the meaning of the term mutation. [1]

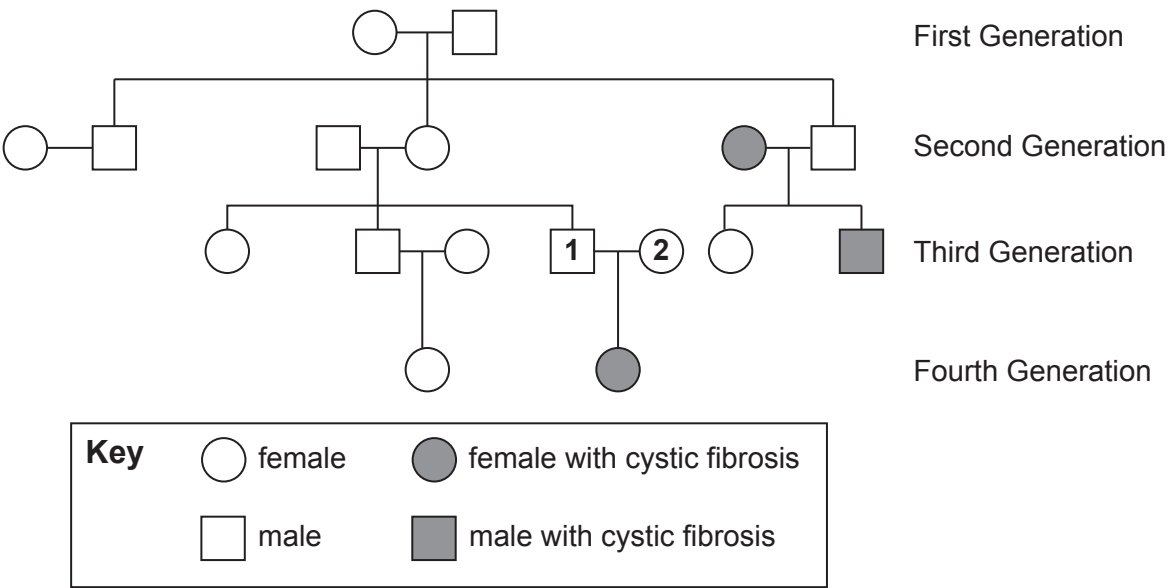
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(b) State an environmental factor that will increase mutation rates. [1]

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(c) The family tree shows the inheritance of cystic fibrosis. Cystic fibrosis is caused by a recessive allele.



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II. State whether Anwen has Type 1 or Type 2 diabetes and explain your answer. [1]

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.....

(ii) State **two** ways in which diabetes can be treated. [2]

1. ....

2. ....

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(ii) State how the population size estimated by the capture/recapture technique compares to the pest control officer’s initial on-the-spot estimate. [1]

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(iii) State **two** ways in which the accuracy of the estimated population size obtained by the capture/recapture technique could be improved. [2]

1. ....

2. ....

(b) State **three** assumptions that must be made when using capture/recapture data. [3]

1. ....

2. ....

3. ....

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END OF PAPER

